

SOVAN PAL

MSc in Data Science and AI, RKMVERI, Belur

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SUMMARY

My background combines a strong foundation in Machine Learning, Deep Learning, and AI concepts with hands-on experience in building intelligent and data-driven applications. I am comfortable working on real-world projects involving NLP, Reinforcement Learning, and distributed AI systems, with a passion for learning emerging technologies and developing impactful solutions.

EDUCATION

Presidency University, Kolkata

BSc (Honours) in Mathematics, 2021–24

CGPA – 7.22

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)

MSc in Data Science and Artificial Intelligence, 2025–27

CGPA – 7.67(till Sem 1)

EXPERIENCE

Research Assistant

May'26 – Present

Flame University

Researching recommender systems using collaborative filtering, NLP, and machine learning techniques. Developed data processing pipelines, implemented TF-IDF and BERT-based feature extraction, and evaluated large-scale e-commerce review datasets for recommendation and prediction tasks.

SKILLS

- **Languages:** Python, C, R Programming, LaTeX
- **Frameworks:** Scikit-Learn, NumPy, Pandas, Streamlit, PyTorch, TensorFlow, Matplotlib
- **Operating Systems:** Windows, Ubuntu Linux
- **Tools:** Git/GitHub, Jupyter Notebook, Google Colab, MS Office

COURSE WORK

- Machine Learning, Artificial Intelligence, Deep Learning, LLM, Agentic AI, Blockchain
- Probability and Statistics, Linear Algebra, Time Series Analysis
- Distributed Computing, Data Structures and Algorithms

ACHIEVEMENTS

- **INSPIRE** Scholarship (Top 1 % of board performers, Govt. of India)
- **SVMCM** Scholarship (Govt. of West Bengal)

CERTIFICATES

- **Accenture North America – Data Analytics & Visualization Job Simulation** (Forge, 2024)
- **The Joy of Computing using Python** (NPTEL, 2026)

PROJECTS

AI Supply Chain Risk Intelligence System

RKMVERI

Apr 2026 – Ongoing

Developed an AI-powered supply chain disruption intelligence platform using SPERT, Neo4j Knowledge Graphs, GraphSAGE, FastAPI, and PyTorch. Built an end-to-end pipeline for extracting entities and relationships from disruption news, generating dynamic supply chain knowledge graphs, and performing downstream impact analysis. Implemented a multi-agent AI architecture to compute hybrid graph-based risk scores and provide real-time disruption monitoring and decision support.

Distributed Market Analyst – Multi-Agent Financial Intelligence System

RKMVERI

Mar 2026 – May 2026

Developed a distributed multi-agent financial intelligence system using Ray, FastAPI, and Streamlit for stock market analysis and forecasting. Combined multimodal time-series forecasting, financial news sentiment analysis, technical indicators, and LLM-based decision fusion using Time-VLM, CLIP embeddings, NewsAPI, Yahoo Finance, Alpha Vantage, and Groq LLaMA-3. Designed a scalable multi-agent architecture to generate real-time BUY/SELL/HOLD recommendations and support data-driven investment decisions.

Multimodal Spectral Fusion for Wheat Disease Classification

RKMVERI

Feb 2026 – Apr 2026

Developed a deep learning framework for wheat disease classification using multispectral (MS) and hyperspectral (HS) imagery. Built modality-specific models, implemented MS+HS cross-modal attention fusion, and evaluated learned 256-D fusion embeddings using SVM, KNN, Logistic Regression, and Random Forest. Achieved 74.17% validation accuracy using multimodal spectral fusion while demonstrating that spectral imaging significantly outperforms traditional RGB-based approaches for crop disease detection

Custom Coin Collector using Reinforcement Learning

RKMVERI

Oct 2025 – Dec 2025

Developed a custom reinforcement learning environment where an intelligent agent learns optimal coin-collection strategies through interaction with the environment. Implemented reward optimization, state-action decision making, and policy learning using Python-based RL techniques. Evaluated agent performance across multiple training episodes to maximize cumulative rewards and demonstrate autonomous problem-solving capabilities.

Startup Success Prediction using Machine Learning

RKMVERI

Sep 2025 – Nov 2025

Developed a machine learning-based predictive analytics system to estimate startup success using historical business and financial indicators. Performed data preprocessing, feature engineering, exploratory data analysis, and model training using multiple ML algorithms. Evaluated model performance using classification metrics and identified key factors influencing startup growth, funding potential, and long-term success.